Slide 1

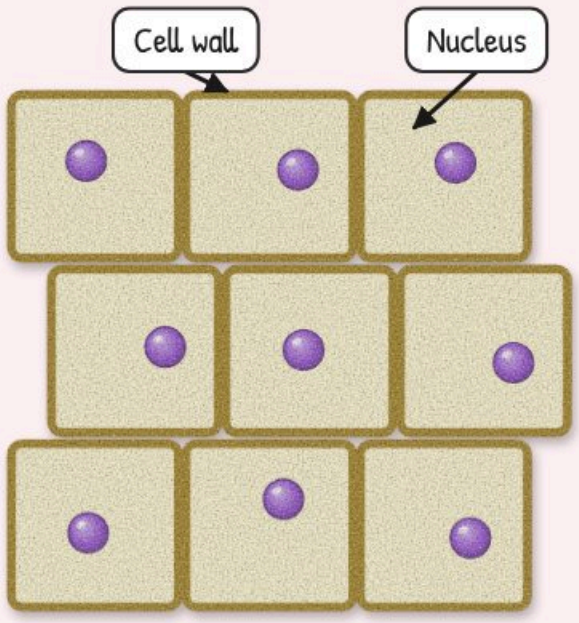
What is a Cell? – Discovery & Cell Theory

Cell: The Unit of Life

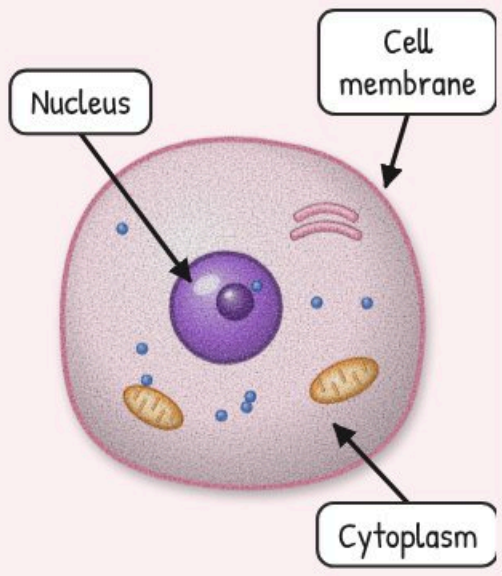
The cell is the fundamental structural and functional unit of all living organisms. Anything less than a complete cell cannot ensure independent living.

Cells under the Microscope

(as seen in the lab)



Onion peel cells

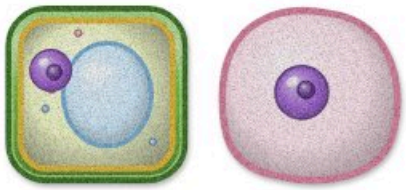


Human cheek cell

Milestones

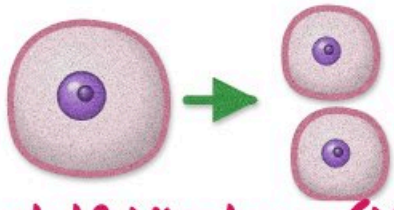
1. **Anton von Leeuwenhoek** – First saw and described a live cell.
2. **Robert Brown** – Later discovered the nucleus.
3. **Schleiden (1838)** – German botanist: all plants are made of cells.
4. **Schwann (1839)** – British zoologist: animal cells have a thin outer layer – the plasma membrane; cell wall is unique to plants.

Cell Theory



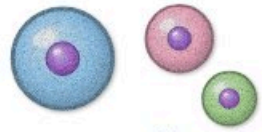
Schleiden & Schwann

Together formulated the cell theory – but did not explain how new cells form.



Rudolf Virchow (1855)

Omnis cellula-e cellula – cells divide and new cells arise from pre-existing cells.



Modern Cell Theory

(i) All living organisms are made of cells and products of cells. (ii) All cells arise from pre-existing cells.

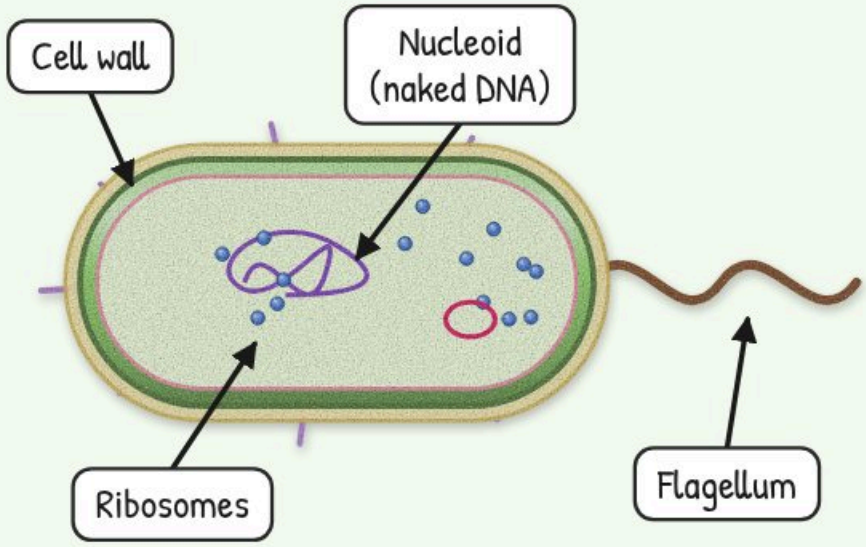
Slide 2

An Overview of Cell – Prokaryotic vs Eukaryotic

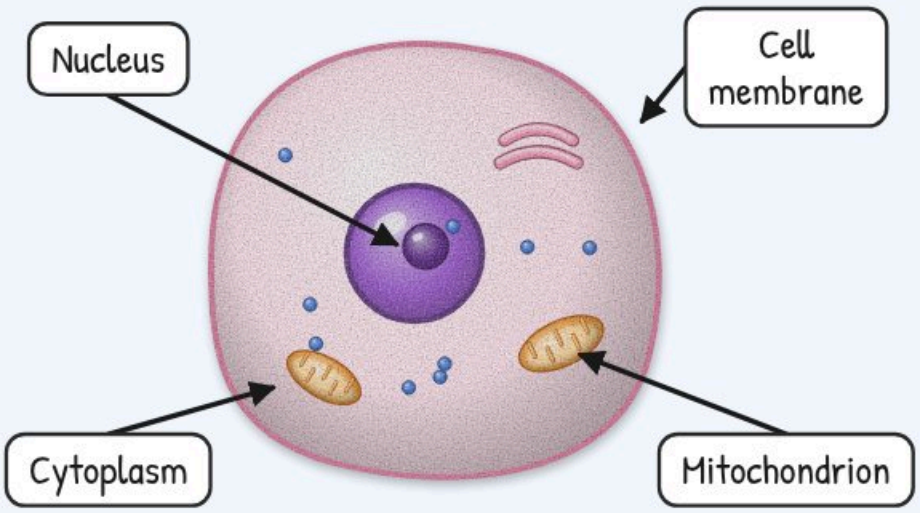
Cell: The Unit of Life

Cells without a membrane-bound nucleus are prokaryotic. Cells with a membrane-bound nucleus and membrane-bound organelles are eukaryotic.

Prokaryotic Cell
(e.g. bacterium)



Eukaryotic Cell
(e.g. animal cell)



Key Differences

Feature	Prokaryotic	Eukaryotic
Nucleus	No nuclear membrane – nucleoid	True nucleus with nuclear envelope
Organelles	No membrane-bound organelles	ER, Golgi, mitochondria, plastids...
Ribosomes	70S	80S (70S in mitochondria & plastids)
Size	Smaller; multiply rapidly	Generally larger
Examples	Bacteria, blue-green algae, mycoplasma, PPLO	Protists, fungi, plants, animals

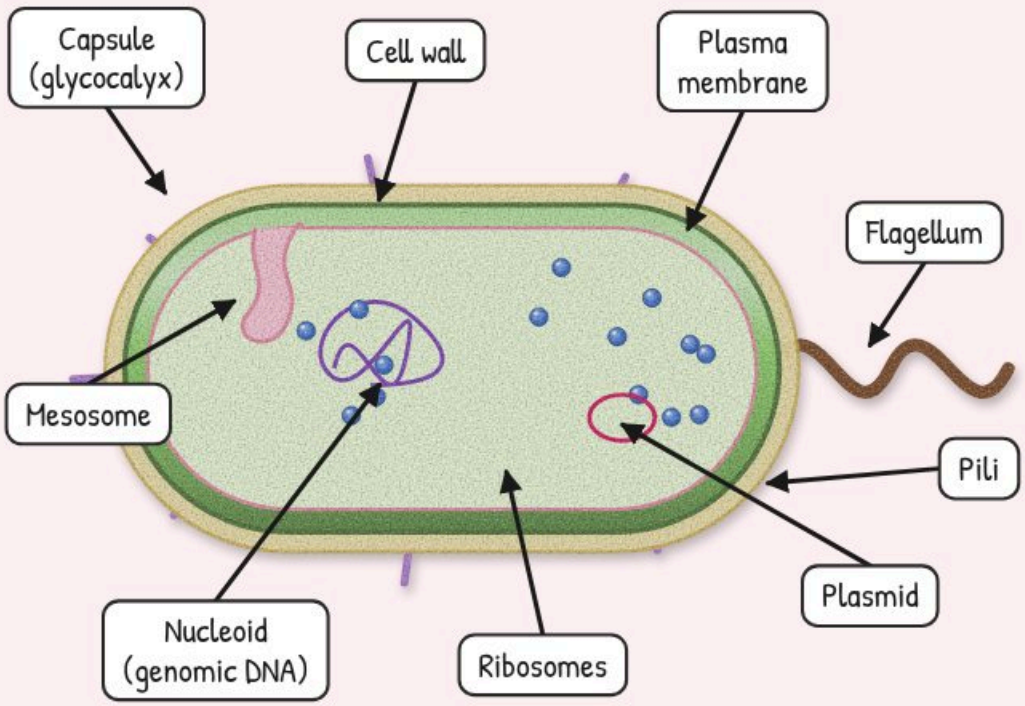
Slide 3

Prokaryotic Cell – Structure of a Bacterium

Cell: The Unit of Life

Prokaryotes have a cell envelope and no membrane-bound organelles. Genetic material is naked and lies in the nucleoid.

Structure of a Bacterial Cell
(diagrammatic view)



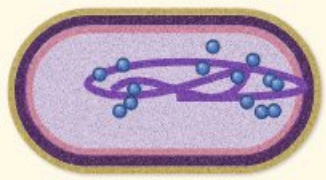
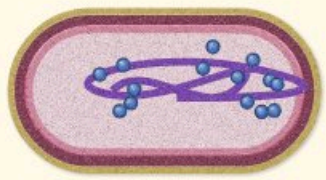
Key Features

1. **Genomic DNA** – Single, circular chromosome; not enveloped by a nuclear membrane.
2. **Plasmids** – Small circular DNA outside genomic DNA; give traits like antibiotic resistance.
3. **Mesosome** – Infolding of plasma membrane – helps in cell wall formation, DNA replication, respiration & secretion.
4. **Chromatophores** – Membranous extensions in cyanobacteria that contain pigments.

Cell Envelope – 3 Layers

1. **Glycocalyx (outermost)** – Loose sheath = slime layer; thick & tough = capsule.
2. **Cell wall** – Determines the shape and prevents the cell from bursting.
3. **Plasma membrane** – Selectively permeable; similar in structure to eukaryotes.

Gram Staining

Gram +ve	Gram –ve
	
Take up the stain	Do not take it up

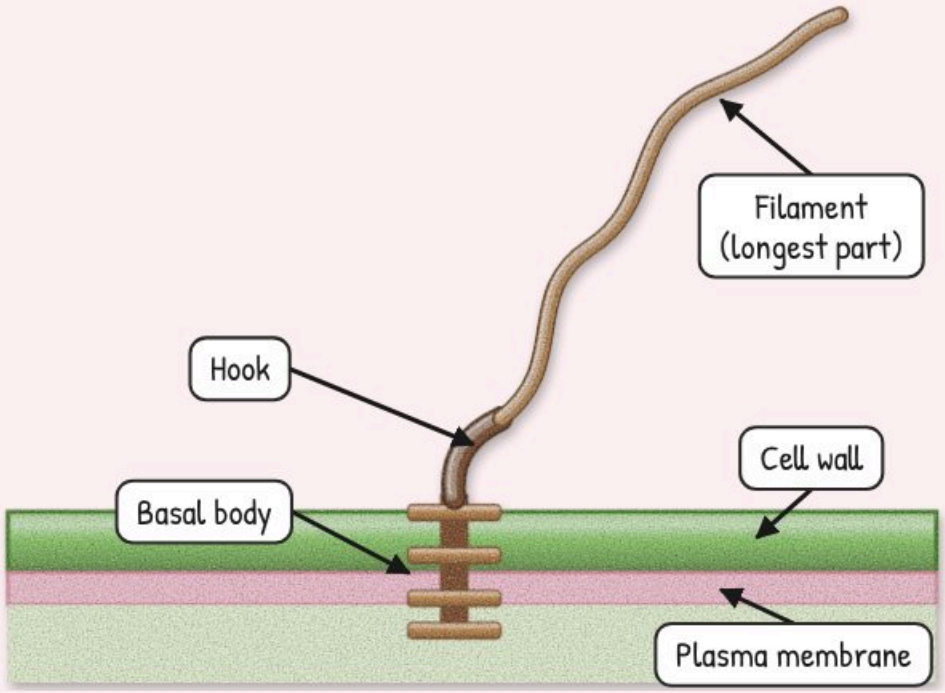
Slide 4

Prokaryotic Cell – Appendages, Ribosomes & Inclusions

Cell: The Unit of Life

Bacteria bear surface structures for movement and attachment, 70S ribosomes for protein synthesis and stored reserve material in the cytoplasm.

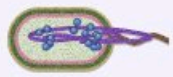
Bacterial Flagellum (3 parts)



Surface Structures

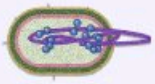
Flagella

1. Thin filamentous extensions for motility; filament, hook & basal body.



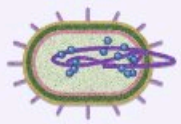
Pili

2. Elongated tubular structures made of a special protein.



Fimbriae

3. Small bristle-like fibres; help attach to rocks in streams & to host tissues.

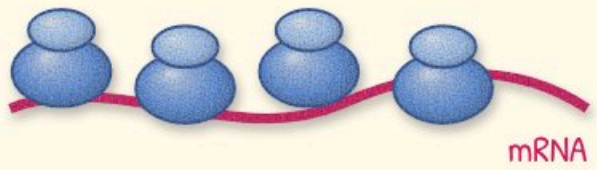


Ribosomes & Polysome



70S ribosome

translate mRNA into proteins



Polysome = many ribosomes on one mRNA

Inclusion Bodies

1. **Reserve material**
   Lie free in cytoplasm; not bounded by any membrane.
2. **Examples**
   Phosphate granules, cyanophycean granules & glycogen granules.
3. **Gas vacuoles**
   In blue-green, purple & green photosynthetic bacteria.

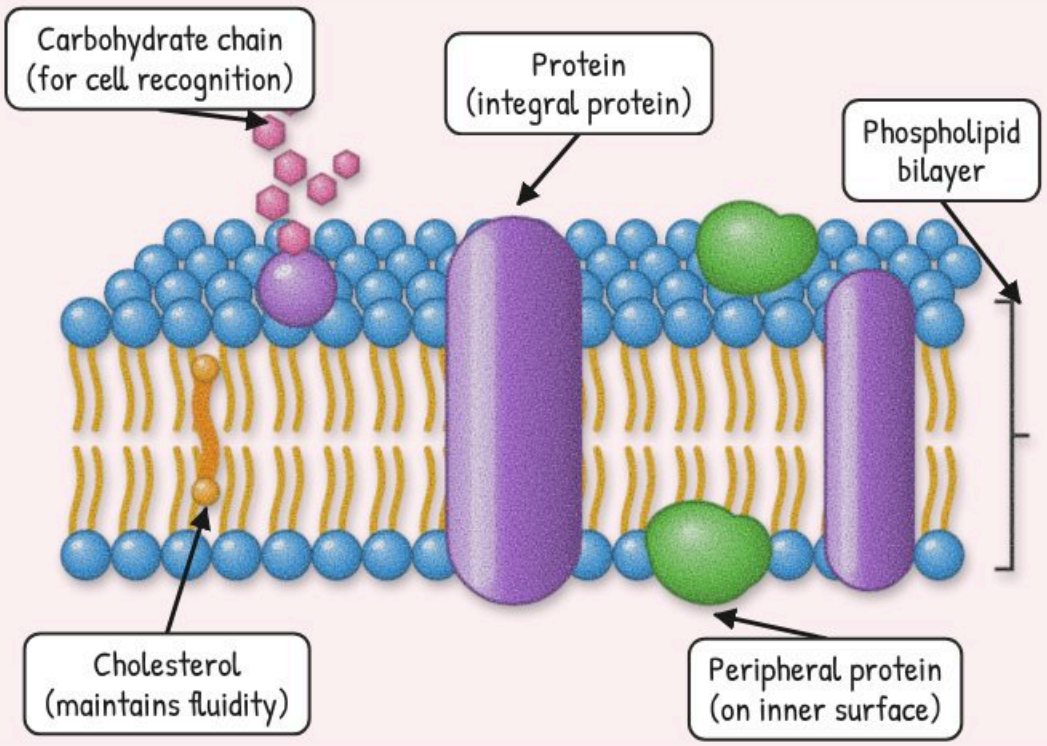
Slide 5

Cell Membrane – Structure & Transport

Cell: The Unit of Life

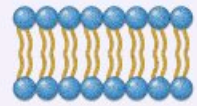
The cell membrane (plasma membrane) is a thin, flexible and living membrane that surrounds the cytoplasm and is selectively permeable.

Structure of Cell Membrane
(Fluid Mosaic Model)

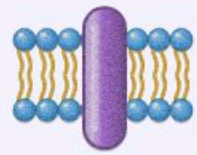


Key Features

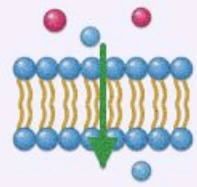
1 Fluid mosaic model
Given by Singer & Nicolson (1972). Quasi-fluid lipid lets proteins move sideways.



2 Lipids + proteins
Phospholipids in a bilayer; in human RBC: 52% protein, 40% lipids.

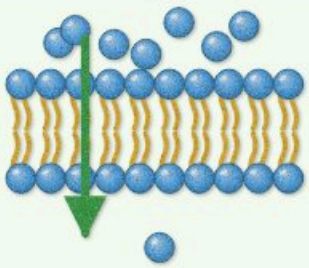


3 Selectively permeable
Allows only certain substances to enter and leave the cell.

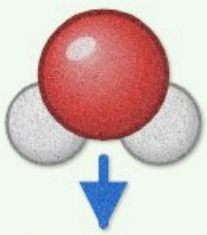


Movement Across the Membrane

Passive – no energy needed

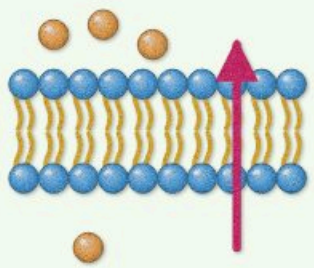


Diffusion
neutral solutes, high → low

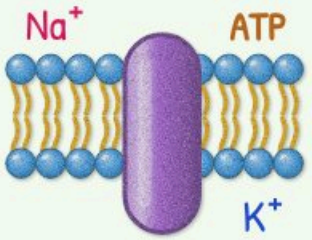


Osmosis
movement of water

Active – uses energy (ATP)



Against gradient
low → high concentration



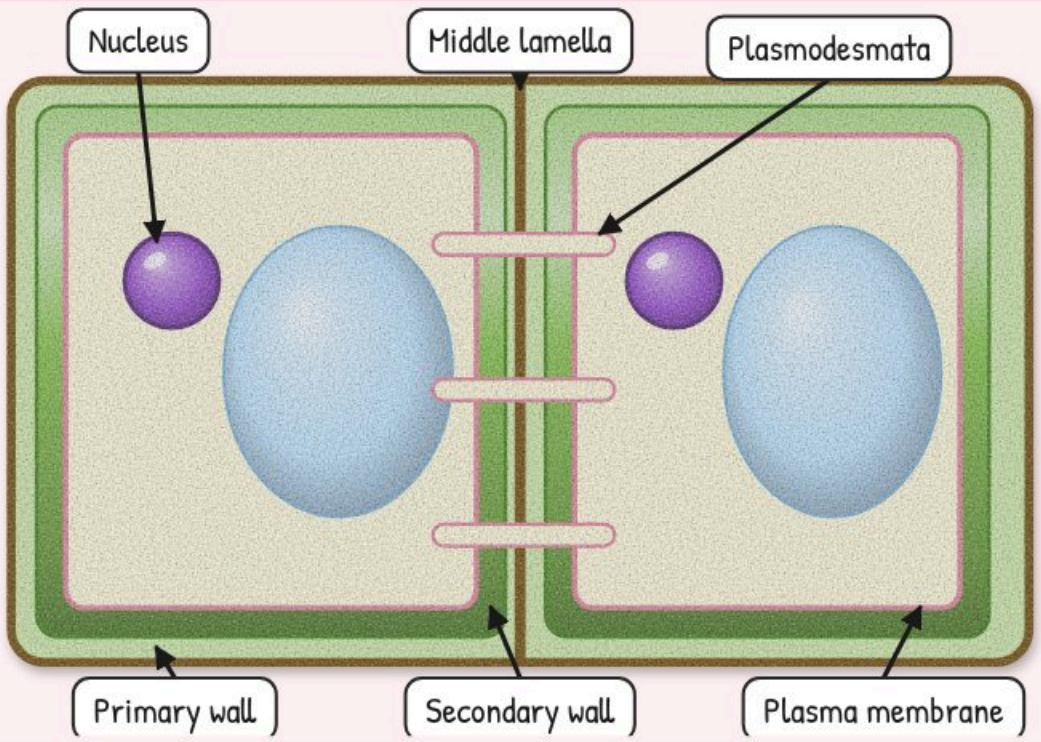
Na^+/K^+ pump
e.g. ions pumped by ATP

Slide 6

Cell Wall –
Structure & Functions

A non-living rigid structure forming the outer covering of the plasma membrane of fungi and plants.

Two Neighbouring Plant Cells
(diagrammatic view)



Composition

In algae
1. Cellulose, galactans, mannans and minerals like calcium carbonate.

In plants
2. Cellulose, hemicellulose, pectins and proteins.

Primary wall
3. Young wall; capable of growth; diminishes as the cell matures.

Secondary wall
4. Formed on the inner (towards membrane) side of the cell.

Functions of Cell Wall

1. Gives shape to the cell.
2. Protects from mechanical damage and infection.
3. Helps in cell-to-cell interaction.
4. Barrier to undesirable macromolecules.

Joining the Cells

Middle lamella
Layer of calcium pectate that holds (glues) neighbouring cells together.

Plasmodesmata
Strands that cross the wall & middle lamella, connecting the cytoplasm of neighbouring cells.

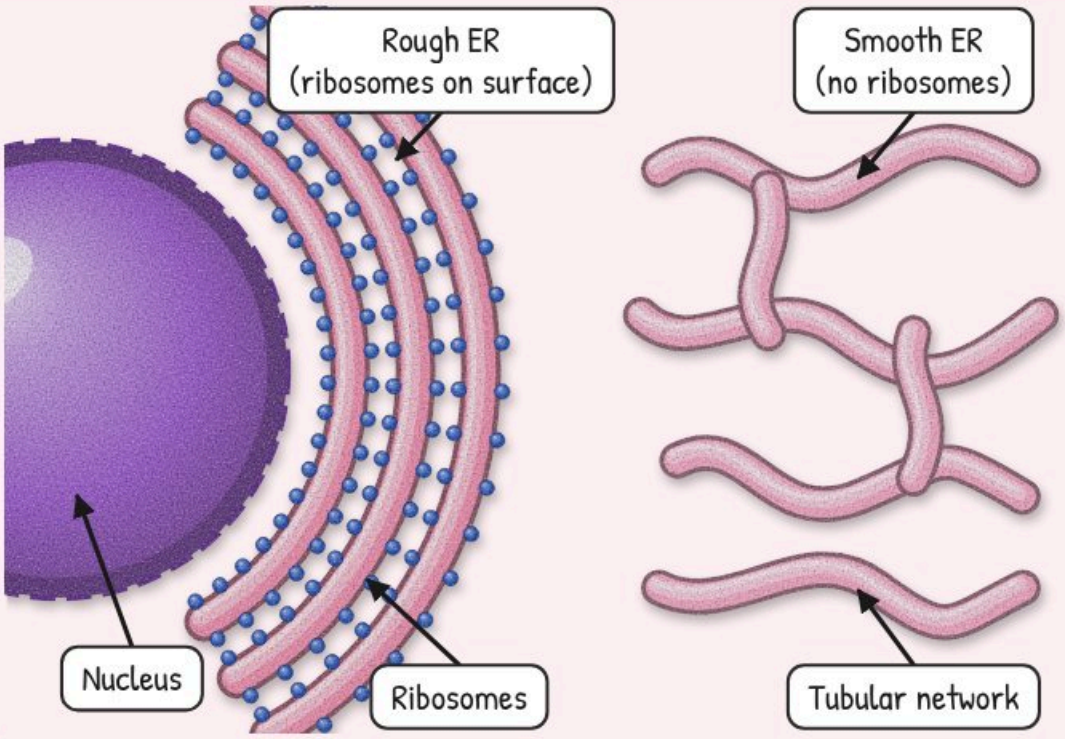
Slide 7

Endoplasmic Reticulum – The Endomembrane System

Cell: The Unit of Life

ER, Golgi complex, lysosomes and vacuoles work in a coordinated way – the endomembrane system. Mitochondria, chloroplasts & peroxisomes are not part of it.

Rough & Smooth ER
(diagrammatic view)



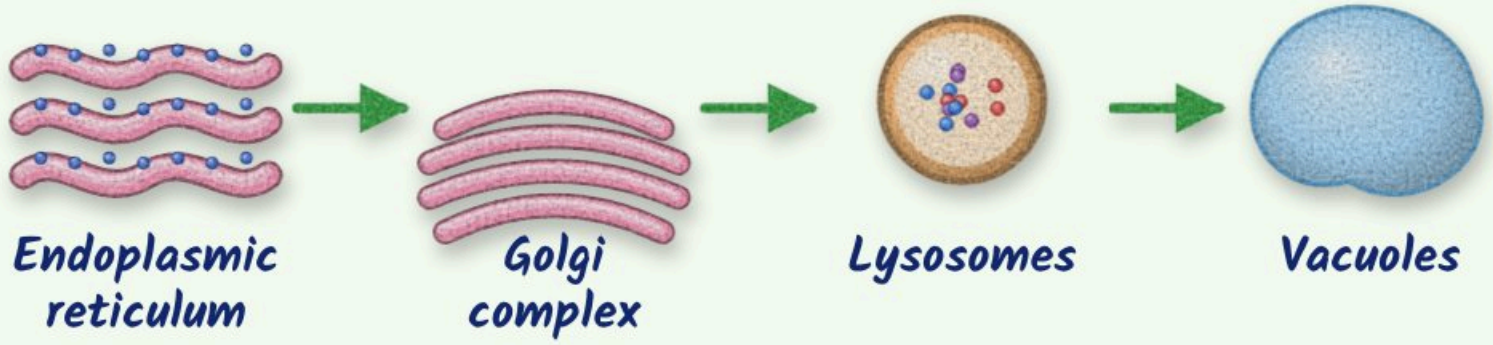
Key Features

1. **Network of tubules**
   Divides the space into luminal (inside ER) and extra-luminal (cytoplasm).

2. **Rough ER (RER)**
   Bears ribosomes; active in protein synthesis & secretion; continuous with outer nuclear membrane.

3. **Smooth ER (SER)**
   Major site of lipid synthesis; in animal cells makes steroidal hormones.

Members of the Endomembrane System



NOT included

Mitochondria
Chloroplasts
Peroxisomes

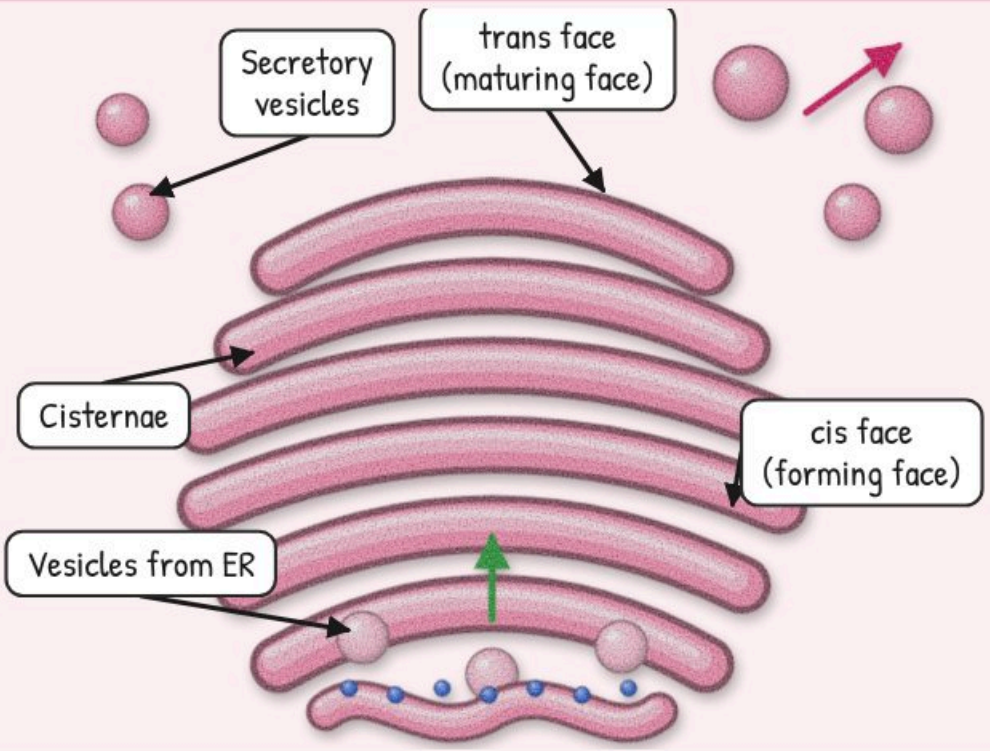
Slide 8

Golgi Apparatus – Structure & Function

Cell: The Unit of Life

Densely stained reticular structures near the nucleus, first observed by Camillo Golgi (1898). Made of flat, disc-shaped sacs called cisternae.

Structure of Golgi Complex
(diagrammatic view)



Key Features

1 Cisternae
Stacked parallel; 0.5 µm to 1.0 µm in diameter.

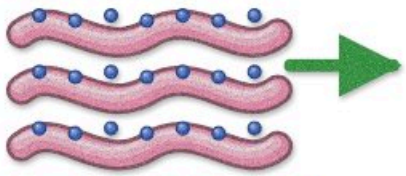
2 Two faces
cis (forming) and trans (maturing) – different but interconnected.

3 Works with ER
Materials from ER fuse with the cis face and move towards the trans face.

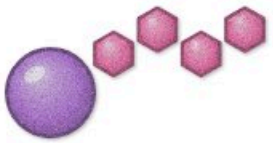
Functions of Golgi Apparatus



Packaging
Packs materials for intracellular targets or secretion outside the cell.



Modification
Proteins made by ribosomes on ER are modified in the cisternae.



Glycoproteins & glycolipids
Golgi is the important site of their formation.

Slide 9

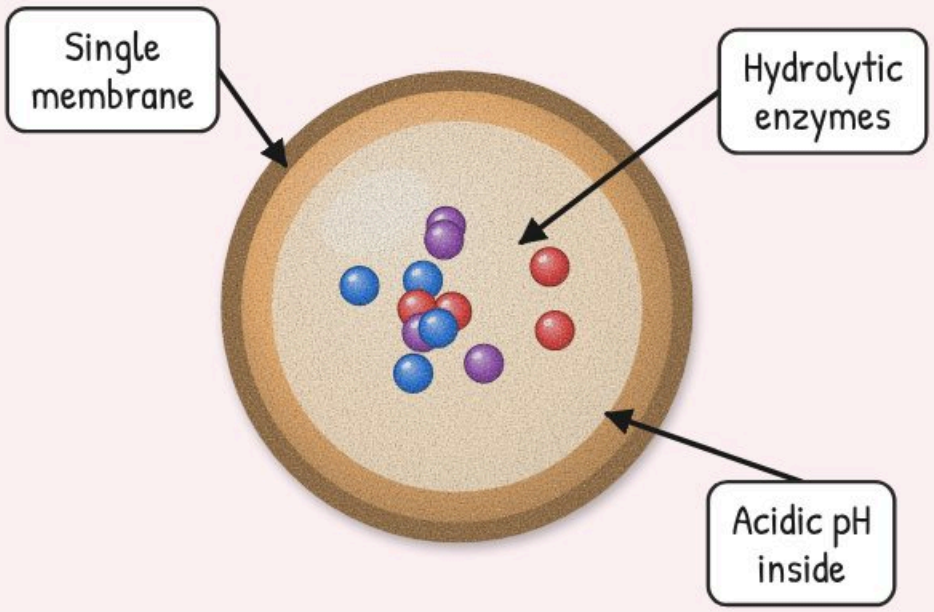
Lysosomes & Vacuoles

Structure & Function

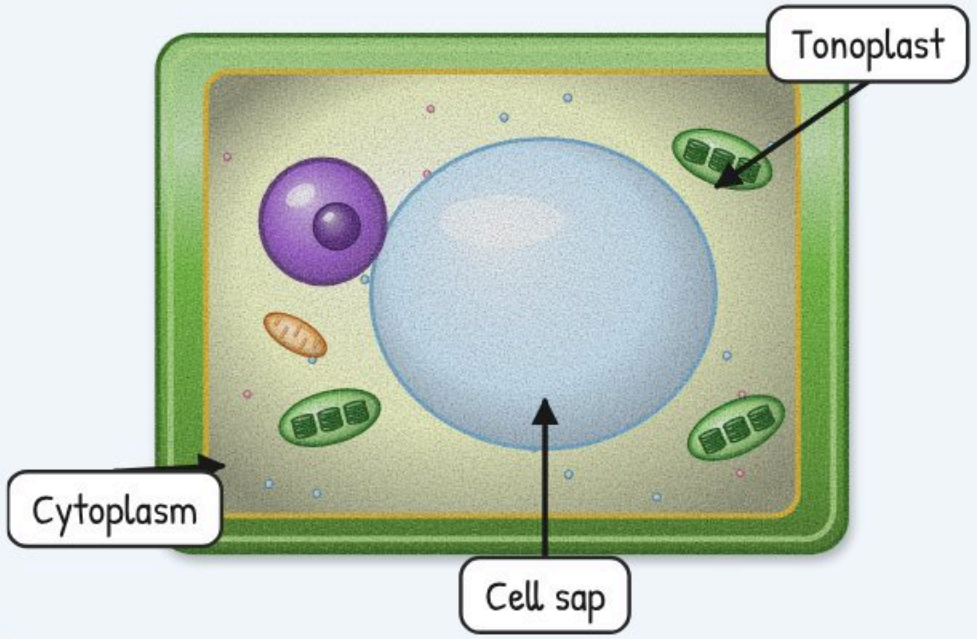
Cell: The Unit of Life

Both are single membrane-bound parts of the endomembrane system – lysosomes digest, vacuoles store.

Lysosome
(suicidal bag)



Vacuole in a Plant Cell



Lysosomes

1. **Formation**
   Membrane-bound vesicles formed by packaging in Golgi.
2. **Enzymes**
   Hydrolases – lipases, proteases, carbohydrases – active at acidic pH.
3. **Function**
   Digest carbohydrates, proteins, lipids & nucleic acids.

Vacuoles

1. **Tonoplast**
   Single membrane; transports ions against gradient into the vacuole.
2. **Size**
   May occupy up to 90% of a plant cell volume.
3. **Special types**
   Contractile vacuole in Amoeba – osmoregulation & excretion; food vacuoles in protists.

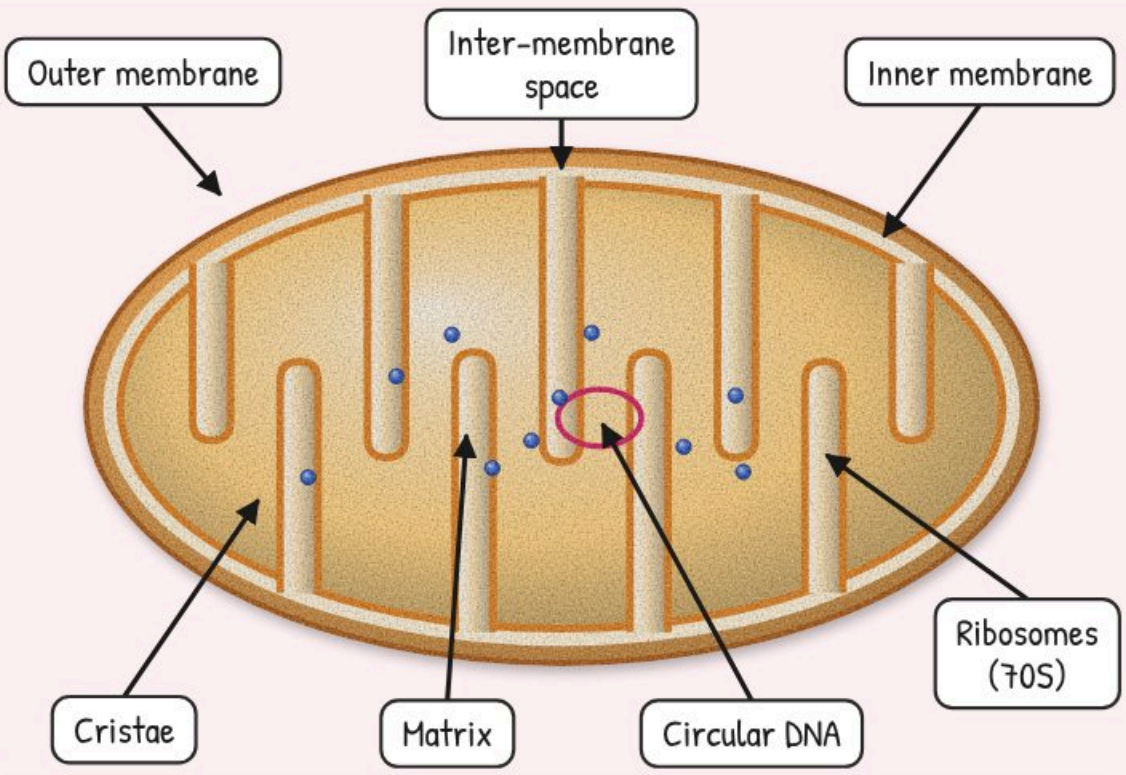
Slide 10

Mitochondria –
The Powerhouse of the Cell

Cell: The Unit of Life

Double membrane-bound organelles; sites of aerobic respiration that produce cellular energy as ATP.

Structure of Mitochondrion
(longitudinal section)



Key Features

1. **Shape & size**
   Sausage-shaped or cylindrical; 0.2–1.0 µm wide, 1.0–4.1 µm long.
2. **Two compartments**
   Outer (inter-membrane space) and inner (matrix).
3. **Cristae**
   Infoldings of inner membrane that increase surface area.
4. **Number**
   Varies with the physiological activity of the cell.

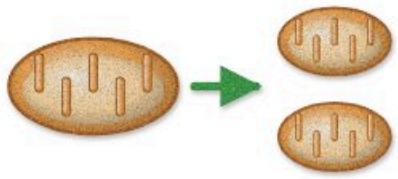
Why Mitochondria are Special



Powerhouse
Sites of aerobic respiration; produce energy as ATP.



Own DNA & ribosomes
Matrix has a single circular DNA, a few RNA and 70S ribosomes.



Divide by fission
Make their own proteins & multiply by fission – semi-autonomous.

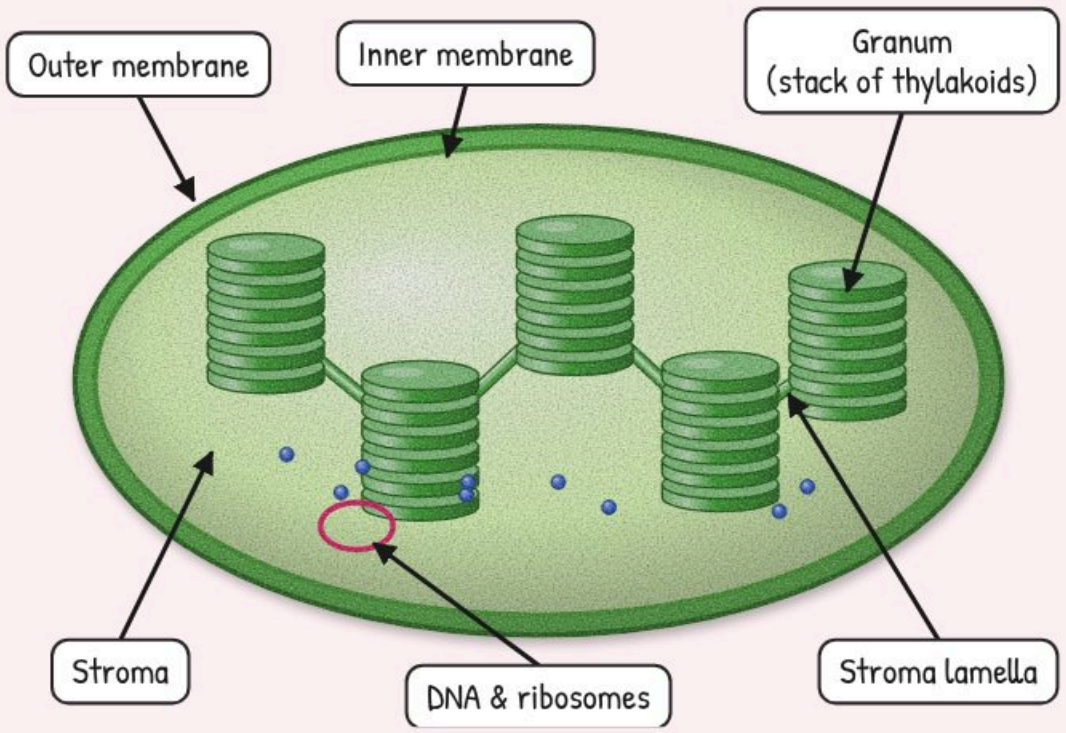
Slide 11

Plastids –
Chloroplast & Types of Plastids

Cell: The Unit of Life

Found in all plant cells and in euglenoids. They are large, bear specific pigments and give colour to plants.

Structure of Chloroplast
(diagrammatic view)



Key Features

1. **Location**
   Mainly in mesophyll cells of leaves.
2. **Shape & size**
   Lens-shaped, oval, spherical or ribbon-like; 5–10 µm long, 2–4 µm wide.
3. **Number**
   1 per cell (Chlamydomonas) to 20–40 per cell (mesophyll).
4. **Stroma**
   Has enzymes for sugar & protein synthesis, circular DNA and 70S ribosomes.

Types of Plastids



Chloroplasts
Chlorophyll & carotenoids; trap light for photosynthesis.



Chromoplasts
Fat-soluble carotenoids – carotene, xanthophylls: yellow, orange, red.



Leucoplasts
Colourless; store food: amyloplasts – starch, elaioplasts – oils & fats, aleuroplasts – proteins.

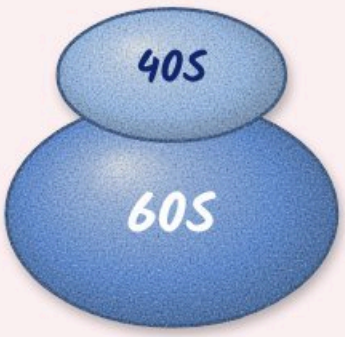
Slide 12

Ribosomes & Cytoskeleton
Protein Factories & Cell Skeleton

Cell: The Unit of Life

Ribosomes are granular structures first observed by George Palade (1953), made of RNA and proteins and not surrounded by any membrane.

Types of Ribosomes
(S = Svedberg unit)



80S ribosome
Eukaryotes (cytoplasm)



70S ribosome
Prokaryotes, mitochondria & plastids

Key Features

1. **Two subunits**
   80S = 60S + 40S; 70S = 50S + 30S.
2. **S (Svedberg unit)**
   Sedimentation coefficient – indirectly a measure of density and size.
3. **Function**
   Sites of protein synthesis (translation).

Cytoskeleton – the Cell's Skeleton
network of filamentous proteins in the cytoplasm



Microtubules
Hollow tubes of tubulin.



Microfilaments
Thin filaments of actin.



Intermediate filaments
Rope-like protein fibres.



Functions
Mechanical support, motility & maintaining cell shape.

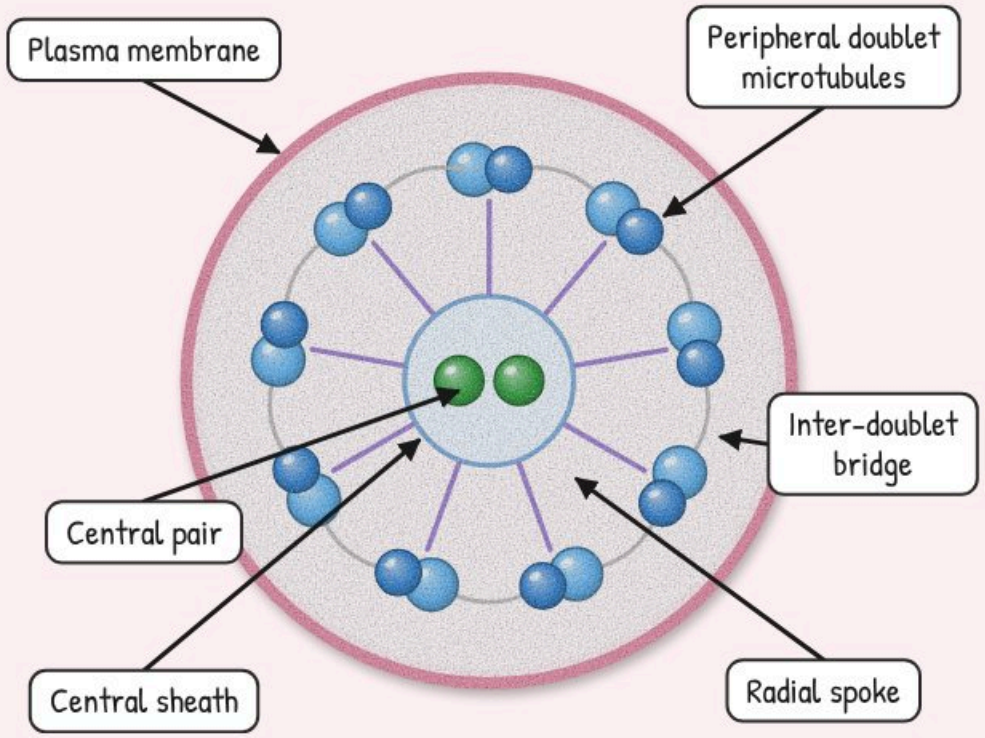
Slide 13

Cilia, Flagella & Centrosome
The 9 + 2 and 9 + 0 Patterns

Cell: The Unit of Life

Cilia and flagella are **hair-like outgrowths** of the cell membrane covered by plasma membrane; their core is the **axoneme**.

Cross Section of Cilium / Flagellum
(9 + 2 array)



Key Features

Cilia

1. Small structures that work like **oars** – move the cell or the fluid around it.

Flagella

2. Comparatively longer; responsible for cell movement.

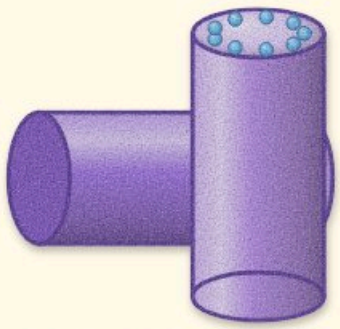
9 + 2 axoneme

3. 9 pairs of **doublet** microtubules around 2 central singlets.

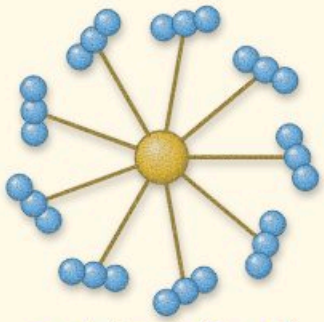
Basal body

4. Centriole-like structure from which cilia & flagella emerge.

Centrosome
(two centrioles)



Perpendicular pair



Cartwheel (9 + 0)

Centrioles

1. Two cylindrical centrioles lie **perpendicular** to each other.
2. 9 evenly spaced triplets of tubulin around a central **hub** – cartwheel.
3. Form basal bodies of cilia/flagella & **spindle fibres** in animal cells.

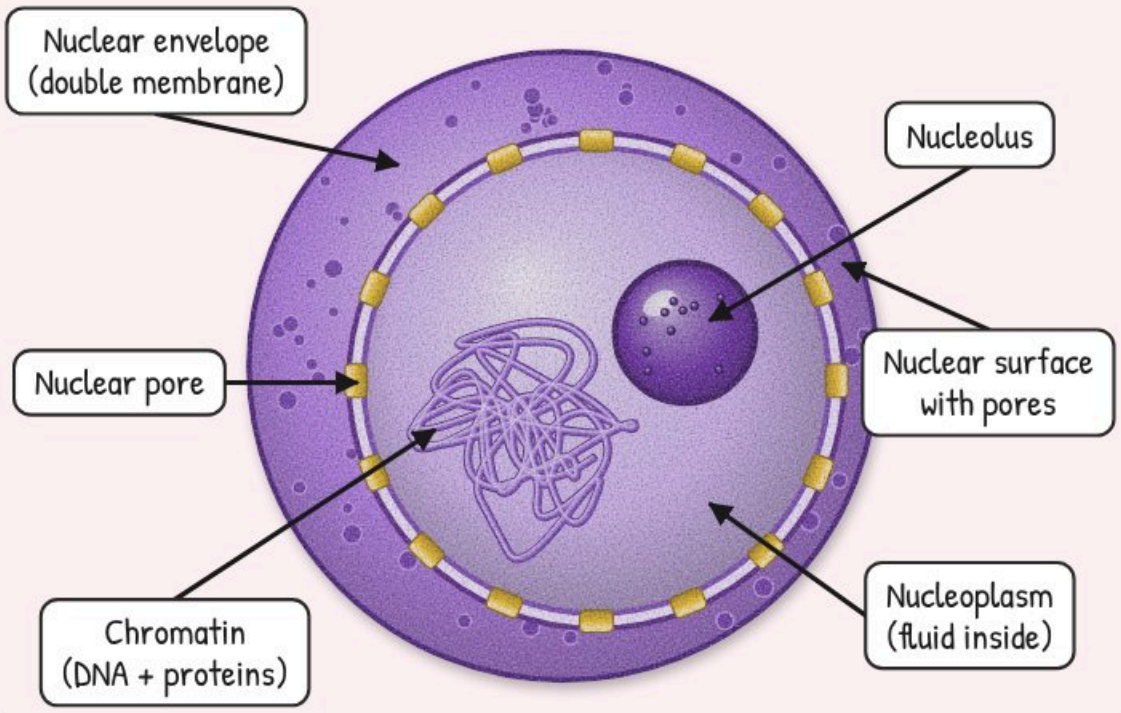
Slide 14

Nucleus –
Structure & Function

Cell: The Unit of Life

First described by Robert Brown (1831). The nucleus contains the genetic material (DNA) and controls the activities of the cell.

Structure of Nucleus
(diagrammatic view)



Key Features

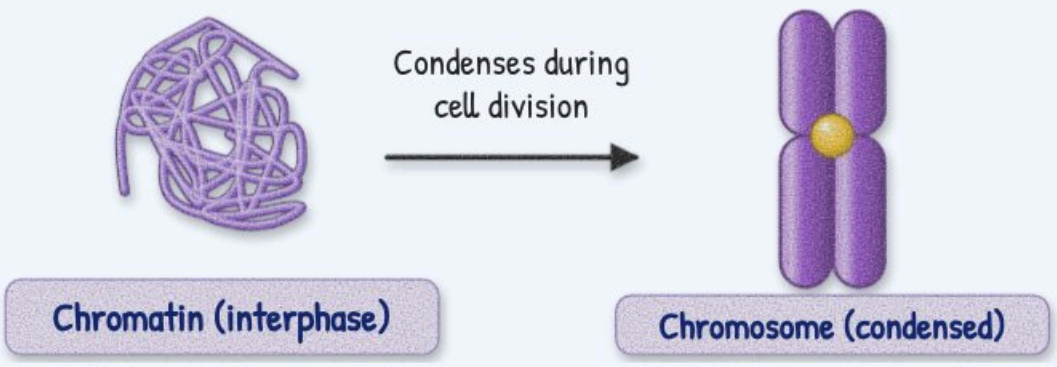
Nuclear envelope
1. Two membranes with a perinuclear space (10–50 nm) between them.

Nuclear pores
2. Passage for RNA and proteins in both directions.

Nucleolus
3. Non-membrane bound; site of active rRNA synthesis.

No nucleus in
4. Mammalian RBCs (erythrocytes) & sieve tube cells of plants.

Chromatin and Chromosomes



Functions of Nucleus

1. Controls all activities of the cell.
2. Stores and transmits genetic information (DNA).
3. Nucleolus makes ribosomal RNA (rRNA).

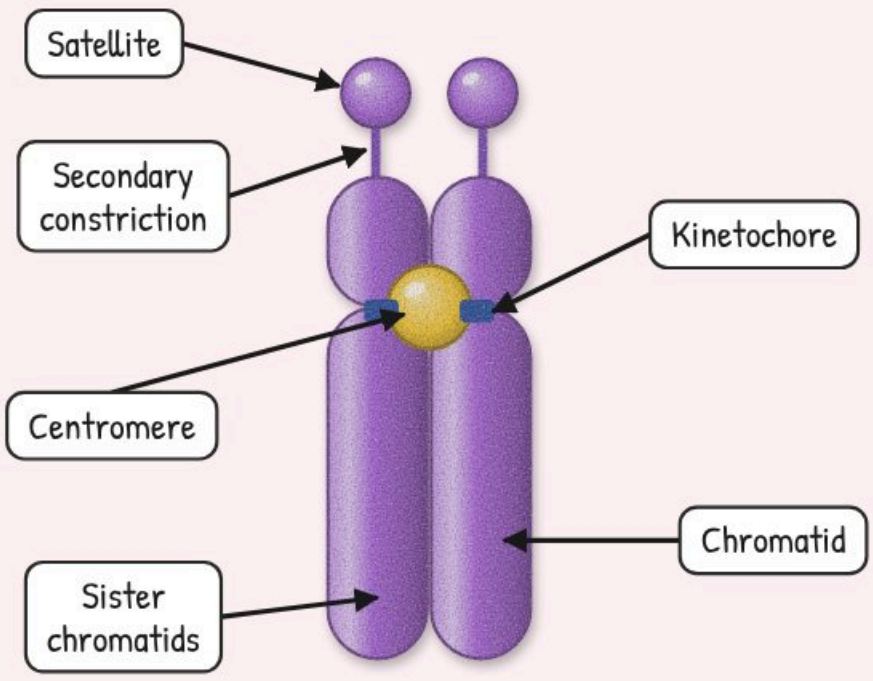
Slide 15

Chromosomes – Types & Microbodies

Cell: The Unit of Life

Every chromosome has a primary constriction, the centromere, with disc-shaped kinetochores on its sides. Its position decides the chromosome type.

Parts of a Chromosome



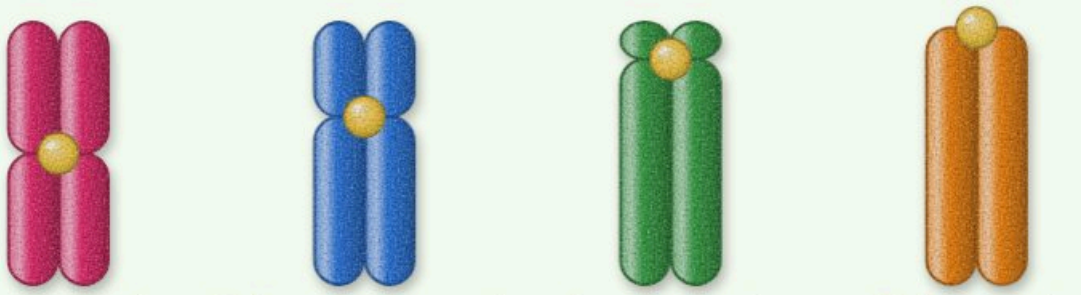
Key Features

Chromatid
1. Each chromosome has two sister chromatids joined at the centromere.

Kinetochores
2. Disc-shaped structures on the sides of the centromere.

Satellite
3. A few chromosomes have a non-staining secondary constriction that gives a small fragment – the satellite.

Types Based on Centromere Position



- **Metacentric** — middle
- **Sub-metacentric** — slightly off
- **Acrocentric** — near one end
- **Telocentric** — at the tip

Microbodies

1. Many membrane-bound minute vesicles with various enzymes.
2. Present in both plant and animal cells.